

Full Platform Engineering Summary

Architecture · Services · Data Flow · Known Issues · Monitoring

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1. Platform Overview

mergers.news is a fully automated M&A; intelligence platform that ingests merger and acquisition filings from global regulatory sources, enriches them using AI extraction, deduplicates records, and publishes them to a static frontend served via Vercel. The system is split across four GitHub repositories, each with a distinct role.

Repository	Role	Runtime	Status
mawire-db	Source of truth, deploy controller	GitHub Actions	✓ Active
mawire-platform	Backend ingestion pipeline	Railway (Node.js)	✓ Active
mawire-site	Static frontend	Vercel (CDN)	✓ Active
mawire-monitor	Independent health auditor	GitHub Actions	✓ Active

1.1 Data Flow

Data enters the platform through five parallel ingestor services running on mawire-platform. All ingestors write to a shared PostgreSQL database (Neon serverless). Every two hours the export script dumps the full deals dataset to deals.json in mawire-db, which Vercel deploys to the CDN. The frontend fetches this file at page load. Separately, mawire-monitor audits the database every hour and publishes a health snapshot to mawire-site for the dashboard.

```
SEC EDGAR / GDELT / EU Registry / APAC Exchanges / News RSS
    ↓ (ingested every 30 min - 12 hours)
mawire-platform → PostgreSQL (Neon)
    ↓ (every 2 hours)
export-to-github.js → mawire-db/deals.json → mawire-site CDN
    ↓ (every 1 hour)
mawire-monitor → pie-health.json → mergers.news/pie-health.json
```

2. mawire-db — Source of Truth

All application source code lives in mawire-db. No service runs directly from this repo; instead, `deploy.py` copies files to the appropriate target repos whenever a push is made to `main` or a `claude/**` branch.

2.1 `deploy.py`

Copies `FIX-*` and `DEPLOY-*` files to `mawire-platform` and `mawire-site` via the GitHub Contents API. `mawire-site` failures are fatal (exit 1); `mawire-platform` failures are warnings only. SHA-aware: reads existing file SHA before PUT to avoid conflicts. Retries on 401/403/409/429 with exponential backoff (2s, 4s, 8s).

2.2 GitHub Actions Workflows

Workflow	Trigger	Purpose
<code>deploy.yml</code>	Push to <code>main/claude/**</code>	Runs <code>deploy.py</code> — copies all <code>FIX-/DEPLOY-</code> files to target repos
<code>auto-backfill.yml</code>	Schedule: every 30 min	Runs <code>FIX-historical-backfill.js</code> (1993–2024), then exports <code>deals.json</code>
<code>refresh-deals.yml</code>	Schedule (periodic)	Re-exports <code>deals.json</code> from DB to <code>mawire-db</code>
<code>historical-backfill.yml</code>	<code>workflow_dispatch</code> (manual)	One-shot full historical backfill with configurable year range
<code>enrich-deals.yml</code>	<code>workflow_dispatch</code> (manual)	Runs <code>FIX-enrich-deals.js</code> on pending deals

2.3 Key Source Files

File	Deploys To	Description
<code>FIX-scheduler-v2.js</code>	<code>mawire-platform/scheduler.js</code>	Master cron orchestrator
<code>FIX-sec-ingestor-index.js</code>	<code>mawire-platform/services/sec-ingestor/index.js</code>	SEC EDGAR filing ingestor
<code>FIX-news-ingestor.js</code>	<code>mawire-platform/services/news-ingestor/index.js</code>	RSS news ingestor
<code>FIX-eu-ingestor.js</code>	<code>mawire-platform/services/eu-ingestor/index.js</code>	EU Merger Registry ingestor
<code>FIX-apac-ingestor.js</code>	<code>mawire-platform/services/apac-ingestor/index.js</code>	HKEX/ASX/SGX ingestor
<code>FIX-export-to-github.js</code>	<code>mawire-platform/scripts/export-to-github.js</code>	DB → <code>deals.json</code> export
<code>FIX-historical-backfill.js</code>	<code>mawire-platform/scripts/historical-backfill.js</code>	Full SEC backfill 1993–present
<code>FIX-enrich-deals.js</code>	<code>mawire-platform/scripts/enrich-deals.js</code>	Deal enrichment script
<code>DEPLOY-index.html</code>	<code>mawire-site/index.html</code>	Main deal listings page
<code>DEPLOY-vercel.json</code>	<code>mawire-site/vercel.json</code>	Vercel routing + security headers
<code>FIX-sw.js</code>	<code>mawire-site/sw.js</code>	Service worker
<code>app.js</code>	<code>mawire-site/app.js</code>	Frontend application logic
<code>style.css</code>	<code>mawire-site/style.css</code>	All frontend styles
<code>pie-health.json</code>	<code>mawire-site/pie-health.json</code>	Health snapshot (updated hourly by monitor)

2.4 Secrets Required

Secret	Used By	Scope Required
MAWIRE_TOKEN	deploy.py, auto-backfill, all workflows	repo + workflow (GitHub PAT)
DATABASE_URL	auto-backfill, historical-backfill, enrich-deals	Neon PostgreSQL connection string

3. mawire-platform — Backend Pipeline

Node.js server running continuously on Railway. The entry point is scheduler.js which registers all cron jobs using node-cron. Each job uses a flag object to prevent overlapping runs. Services run as separate modules under ./services/ and scripts run under ./scripts/.

3.1 Cron Schedule

Service	Frequency	Source / Target	Flag
SEC EDGAR ingestor	Every 30 min	SEC EDGAR EFTS API → PostgreSQL	flags.sec
GDELT ingestor	Every 1 hour	GDELT API → PostgreSQL	flags.gdelt
News RSS ingestor	Every 2 hours	RSS feeds → PostgreSQL	flags.news
EU Merger Registry	Every 12 hours	EU Commission API → PostgreSQL	flags.eu
APAC (HKEX/ASX/SGX)	Every 4 hours	HK/AU/SG exchange APIs → PostgreSQL	flags.apac
AI Extraction queue	Every 5 min	PostgreSQL (needs_review=true) → update	flags.extract
Deduplication	Every 6 hours	PostgreSQL → merge near-duplicates	flags.dedup
GitHub export	Every 2 hours	PostgreSQL → mawire-db/deals.json	(no flag)
Stats cache clear	Every 1 hour	DELETE FROM stats_cache WHERE key=api_stats	(no flag)

3.2 Startup Sequence

On startup, the scheduler fires staggered initialisation runs to warm up the pipeline:

Delay	Service
2 seconds	SEC EDGAR ingestor
15 seconds	News RSS ingestor
30 seconds	EU Merger Registry ingestor
45 seconds	APAC ingestor
60 seconds	GitHub export (deals.json)

3.3 Database Schema (PostgreSQL / Neon)

Table	Purpose	Key Columns
deals	Core deal records	id, acquirer_id, target_id, headline, deal_type, status, deal_value, sector, region, country
companies	Acquirer/target entities	id, name, normalized_name, cik, sic
filings	SEC filing records	id, deal_id, company_id, filing_type, accession_no, cik, edgar_url, filing_date
deal_sources	Per-deal source records	id, deal_id, source_type, source_name, source_url, raw_content
stats_cache	Key/value cache	key, value, updated_at (includes last_github_export timestamp)

pie_checks	PIE health check results	id, check_name, category, status, score, message, detail (jsonb), checked_at
pie_alerts	Active/resolved alerts	id, alert_key, severity, title, description, first_seen, last_seen, resolved_at, auto_remove
pie_metrics	Time-series metric values	id, metric_name, value, unit, recorded_at
pie_health_snapshots	Hourly health snapshots	id, overall_score, pass_count, warn_count, fail_count, active_alerts, snapshot (jsonb)

3.4 AI Extraction Service

The extraction queue runs every 5 minutes and processes up to 20 deals per cycle where needs_review=true and source_confidence < 0.85. It pulls the filing's raw_content, truncates to 10,000 characters, and calls the AI extractor to populate headline, deal_value, per_share_value, premium_pct, announcement_date, and sector. Confidence is capped at 0.95. After processing, needs_review is set to false.

3.5 Deduplication Logic

Runs every 6 hours. Identifies near-duplicate deal pairs where acquirer and target names match (case-insensitive) and the deals were created within 6 hours of each other, excluding "Undisclosed" entities. Caps at 500 pairs per cycle. The PIE monitor checks dedup health as one of its 13 health checks.

3.6 Known Issues — mawire-platform

Issue	Severity	Detail
acquirer column missing	Low	PIE check_name_quality references deals.acquirer which may not exist — check skips
confidence column missing	Low	PIE check_confidence_distribution references deals.confidence — check skips with s
No ingestor retry logic	Medium	Individual ingestor failures only caught with try/catch — no retry or backoff on transier
Extraction capped at 10k chars	Low	Large filings truncated at 10,000 chars — may miss deal value buried deep in docum
stats_cache proxy fallback	Low	export_freshness check falls back to deals.updated_at when stats_cache has no entr

4. mawire-site — Frontend (mergers.news)

Static site hosted on Vercel. No server-side rendering — all data fetched client-side from GitHub raw CDN (deals.json) or same-origin files (pie-health.json). Vercel handles routing via vercel.json with cleanUrls enabled.

4.1 Pages & Routes

URL	Source File	Description
/	index.html	Main deal listings — loads deals.json from CDN
/ipo	ipo.html + ipo.js	IPO tracker with dynamic loader
/about	about.html	About page
/contact	contact.html	Contact form (Formspree integration)
/tender-offers	tender-offers.html	Tender offer listings
/mergers/technology	mergers/technology.html	Technology sector deal listings
/mergers/healthcare	mergers/healthcare.html	Healthcare sector deal listings
/mergers/financial-services	mergers/financial-services.html	Financial services deal listings
/monitoring	monitoring.html	PIE dashboard — SHA-256 password-gated
/404	404.html	Custom 404 error page
/pie-health.json	pie-health.json	Live health snapshot — no-cache headers
/sw.js	sw.js	Service worker — no-cache headers

4.2 Vercel Configuration (vercel.json)

Setting	Value	Effect
cleanUrls	true	Strips .html extension — /about instead of /about.html
trailingSlash	false	Redirects /about/ → /about
Rewrites	12 rules	/monitoring→monitoring.html, /deal/(.*)→index.html, /ipo→ipo.html, sector pages, legal pages

4.3 Security Headers

Header	Value
Content-Security-Policy	default-src 'none'; script-src 'self' 'unsafe-inline' GTM; style-src 'self' 'unsafe-inline' Google Fonts; con
Strict-Transport-Security	max-age=31536000; includeSubDomains; preload
X-Frame-Options	DENY
X-Content-Type-Options	nosniff
Referrer-Policy	strict-origin-when-cross-origin
Cross-Origin-Opener-Policy	same-origin

Cross-Origin-Resource-Policy	same-origin
Permissions-Policy	camera=(), geolocation=(), microphone=(), payment=(), usb=()
Server	(empty — removes server fingerprint)

4.4 Service Worker (sw.js)

Cache name: mergers-news-v6. Uses a shell-first caching strategy with stale-while-revalidate for static assets. Navigation requests are always network-first with cache fallback. raw.githubusercontent.com requests are network-only with empty array [] fallback to prevent stale CDN 404s from breaking the deals feed. pie-health.json fetches use cache-busting (?_=Date.now()) to always get fresh health data.

Shell URLs pre-cached on install:

- / — index
- /style.css, /app.js, /charts.js, /manifest.json
- /about, /about.js, /ipo, /ipo.js, /contact

4.5 Monitoring Dashboard (/monitoring)

Password-gated via SHA-256 hash in the browser. Session stored in sessionStorage. On unlock, fetches /pie-health.json (same-origin, no CDN caching issues). Displays overall score, category breakdown, per-check status, and active alerts. Password hash: 18af8f28a0a1e4af4f6896e44765996e1afc543614896e4dfa364ef8981c68e0

4.6 Known Issues — mawire-site

Issue	Severity	Detail
unsafe-inline in CSP	Medium	script-src and style-src both allow 'unsafe-inline' — reduces XSS protection. Required
Service worker v6 cache name	Low	If cache version not bumped on deploy, users may see stale assets until SW update c
No offline fallback for deals	Low	If deals.json fetch fails and no cache exists, page shows empty state with no user me
Legal pages referenced in vercel.json but may not exist	Medium	/legal/terms, /legal/privacy, /legal/disclaimer rewrites defined but HTML files not confir

5. mawire-monitor — Platform Integrity Engine

Completely independent auditing system. Runs hourly at :45 via GitHub Actions. Connects directly to the PostgreSQL database, runs 13 health checks, persists results, manages alerts, triggers auto-remediation, and exports pie-health.json. No dependency on mawire-db or mawire-platform — audits those systems from the outside.

5.1 Health Checks — 13 Checks Across 4 Categories

Check Name	Category	Weight	What It Measures	Fail Threshold
ingestion_freshness	ingestion	35%	Age of most recent deal in DB	> 8 hours since last deal
ingestion_velocity	ingestion	35%	Deal count in rolling 24h/7d/30d windows	< 10 deals in last 24 hours
source_diversity	ingestion	35%	Number of active source types in last 90d	< 8 active sources
filing_coverage	ingestion	35%	Presence of all 4 key SEC filing types	> 3 filing types missing
name_quality	quality	30%	Placeholder names (Undisclosed etc.) in active deals	> 30% with placeholder names
value_completeness	quality	30%	Active deals missing deal_value	> 80% missing value
data_completeness	quality	30%	Avg fill rate: sector, region, country, headcount	< 60% average fill
dedup_health	pipeline	20%	Near-duplicate deal pairs in DB	> 50 duplicate pairs
export_freshness	pipeline	20%	Age of last deals.json export	> 14 hours since export
review_queue	pipeline	20%	Deals pending human/AI review	> 200 deals in queue
geographic_coverage	coverage	15%	Distinct regions + deals with region set	N/A (warn only)
sector_coverage	coverage	15%	Distinct sectors in active deals	N/A (warn only)
confidence_distribution	coverage	15%	Avg confidence score + low-confidence deals	< 45% below 0.70

5.2 Scoring Engine

Each check returns a score 0–100. Category score = average of non-skipped checks in that category. Overall score = weighted average of category scores.

Score Range	Status	Meaning
≥ 85	healthy	All systems operating normally
60–84	degraded	One or more checks in warning state — investigate
< 60	critical	One or more checks failing — immediate action required

5.3 Alert Management

Alerts are upserted to pie_alerts table. A failing check creates/updates a critical alert. A warning check creates/updates a warning alert. A passing check resolves any existing alert. Auto-remediation: name_quality failures queue placeholder-name deals for re-enrichment (sets needs_review=true). Dedup alerts are logged but deduplication is handled by the platform scheduler.

5.4 Export

On each run, exports pie-health.json to two repos:

- realfrantheman/mawire-monitor — source of truth, permanent record
- realfrantheman/mawire-site — same-origin serving at mergers.news/pie-health.json

pie-health.json contains: generatedAt, overallScore, status, durationMs, categories (with per-category scores and check details), full checks array, active alerts array, and summary counts (pass/warn/fail/skip/critical/warnings).

5.5 GitHub Actions Workflow

Setting	Value
Schedule	cron: 45 * * * * (every hour at :45)
Trigger	schedule + workflow_dispatch
Runner	ubuntu-latest
Timeout	10 minutes
Permissions	contents: write
Node version	20
Install	npm install (no lockfile — installs pg)
Run	node index.js

5.6 Secrets Required

Secret	Purpose
DATABASE_URL	PostgreSQL connection for all 13 health checks and persisting results
MAWIRE_TOKEN	GitHub PAT for exporting pie-health.json to mawire-monitor and mawire-site

5.7 Known Issues — mawire-monitor

Issue	Severity	Detail
No package-lock.json	Low	npm install resolves latest pg on every run — could break if pg releases breaking change
Node.js 20 deprecation	Low	actions/checkout@v4 and actions/setup-node@v4 will be forced to Node.js 24 after June
acquirer/confidence columns	Low	name_quality and confidence_distribution checks reference columns that may not exist in
No alerting output channel	Medium	Alerts are stored in DB and visible on dashboard only — no email/Slack/webhook notifica

6. Recent Incidents & Fixes

Date (UTC)	Incident	Root Cause	Fix Applied
2026-06-10	/monitoring showing HTTP 404	HEALTH_URL pointed to raw.githubusercontent.com instead of GitHub repo	Changed HEALTH_URL to GitHub repo
2026-06-10	Dashboard showed error even after fix	Double loadHealth() call — one from frontend and one from backend	Removed conditional loadHealth() call. Only scaled R
2026-06-10	Transient 404 window after deploy.py deployed monitoring.html	BEFORE DEPLOYING in deploy.py, push health service	
2026-06-10	Auto-backfill merge conflict on deploy.py	git push failed: refresh-deals was already pushed to main	Replaced refresh-deals with refresh-deals

7. Codex Integration — External Audit System

mawire-monitor is configured as a Codex-controlled external auditing system. It is fully independent from the rest of the platform and includes an AGENTS.md file with complete instructions for OpenAI Codex CLI.

7.1 Repository Setup

File	Purpose
index.js	13 PIE health checks + scoring + alert management + DB persistence + export
package.json	Single dependency: pg ^8.11.0. Scripts: npm start → node index.js
AGENTS.md	Full Codex CLI instructions: how to run, add checks, modify thresholds, SQL debugging queries
.github/workflows/run.yml	GitHub Actions: hourly schedule at :45 + workflow_dispatch for manual trigger

7.2 Codex Access Model

Repository	Read	Write	Actions Trigger
mawire-db	✓	✗	✗
mawire-platform	✓	✗	✗
mawire-site	✓	✗	✗
mawire-monitor	✓	✓ (full)	✓

Codex has full read+write access to mawire-monitor (its home repo) and read-only access to the three operational repos. This allows Codex to modify health checks, update thresholds, commit audit reports, and trigger runs — while having no write access to production code in mawire-db, mawire-platform, or mawire-site.

Implemented via a fine-grained GitHub PAT (CODEX_AUDIT_TOKEN):

- mawire-monitor — Contents: Read+Write, Actions: Read+Write, Metadata: Read
- mawire-db, mawire-platform, mawire-site — Contents: Read, Metadata: Read only

7.3 Audit Capabilities

- Full code scan across all four repositories — bugs, security issues, dead code
- Live website scan at mergers.news — all URLs, HTTP status, asset loading, JS errors
- PIE monitor trigger — run workflow_dispatch and read resulting pie-health.json
- Database query access — cross-reference alerts with live DB state
- Structured markdown audit report with severity classification

7.4 Required Secrets in mawire-monitor

Secret	Value Source	Used For
DATABASE_URL	Neon dashboard	PIE health checks + DB access
MAWIRE_TOKEN	GitHub PAT (repo+workflow)	Export pie-health.json to mawire-monitor + mawire-site

OPENAI_API_KEY	OpenAI dashboard	Codex CLI audit runs
CODEX_AUDIT_TOKEN	GitHub fine-grained PAT	Read-only access to all repos during Codex audit

8. Environment & Deployment Reference

8.1 Infrastructure

Component	Provider	Notes
Backend runtime	Railway	Node.js 18+, always-on, env vars: DATABASE_URL
Database	Neon (serverless PostgreSQL)	SSL required (rejectUnauthorized: false), max 3 pool connections in PIE, conn
Frontend hosting	Vercel	Static site, CDN, automatic HTTPS, custom domain mergers.news
DNS / Domain	mergers.news	Points to Vercel CDN
CI/CD	GitHub Actions	Runs on ubuntu-latest, Node.js 20 (deprecation warning: upgrade before Sep
Code storage	GitHub	All four repos under realfrantheman/
Contact forms	Formspree	CSP allows formspree.io in connect-src and form-action
Analytics	Google Analytics / GTM	CSP allows googletagmanager.com, google-analytics.com, analytics.google.c

8.2 Full Secrets Reference

Secret Name	Repos That Need It	Description
MAWIRE_TOKEN	mawire-db, mawire-monitor	GitHub PAT. Needs repo scope (read/write all repos) + workflow scope (creat
DATABASE_URL	mawire-db, mawire-monitor	Neon PostgreSQL connection string. Format: postgresql://user:pass@host/db
OPENAI_API_KEY	mawire-monitor	OpenAI API key for Codex CLI audit runs (optional until Codex audit workflow
CODEX_AUDIT_TOKEN	mawire-monitor	Fine-grained GitHub PAT: Contents Read on all four repos, Actions Read+W

8.3 Deployment Checklist

- mawire-db: push to main or claude/** branch → deploy.yml triggers automatically
- mawire-platform: deploy.yml copies scheduler.js and all service/script files — Railway auto-redeploys on repo push
- mawire-site: deploy.yml copies all DEPLOY-* and FIX-sw.js, app.js, style.css, vercel.json — Vercel auto-deploys on repo push
- mawire-monitor: self-managed — edit files directly in mawire-monitor repo, GitHub Actions handles the rest
- New secrets: must be added to BOTH mawire-db (for deploy.py) AND target repos independently
- Workflow scope: MAWIRE_TOKEN must have workflow scope to create/modify .github/workflows/ files via API

9. Platform Health Summary

Area	Status	Notes
SEC EDGAR ingestion	✓ Active	Every 30 min, DEFM14A/SC TO-T/S-4/SC 13E-3/DEFA14A/PREM14A
GDELT ingestion	✓ Active	Every 1 hour
News RSS ingestion	✓ Active	Every 2 hours
EU Merger Registry	✓ Active	Every 12 hours
APAC (HKEX/ASX/SGX)	✓ Active	Every 4 hours
Historical backfill	■ Fixed	Merge conflict bug fixed — fetch+reset strategy now used
AI extraction queue	✓ Active	Every 5 min, cap 20 deals/cycle
Deduplication	✓ Active	Every 6 hours
GitHub export (deals.json)	✓ Active	Every 2 hours + on backfill completion
Frontend (mergers.news)	✓ Live	Vercel CDN, full security headers, service worker v6
Monitoring dashboard	✓ Live	/monitoring — SHA-256 password gate, same-origin health fetch
PIE health monitor	✓ Active	Every hour at :45, 13 checks, last run: SUCCESS
Codex audit integration	■ Partial	AGENTS.md and fine-grained PAT setup required — OPENAI_API_KEY not yet a
auto-backfill.yml	✓ Fixed	Merge conflict in deals.json resolved